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SEMESTER - 1

Q1.

Computers have evolved throughout time, and generally there are five generations of computers that are described according to the technology used in each generation and the changes in performance and capabilities over time. Each successive generation has had notable improvements and reduced deficiencies in comparison to the preceding generation.

1st Generation computers (1940-1956):

The first generation of computers was characterized by using vacuum tubes as the major component within the computer. In general, the first generation of computers were enormous, required a significant amount of electricity to operate and produced a large amount of heat because of their size and the volume of digital logic circuits within them. The programming conducted on first-generation computers was conducted using machine code (also known as binary code). The most notable advantages of the first generation of computers was that they provided electronic computation for the first time and could perform calculations more rapidly than humans could. The disadvantages of first generation computers was that they were very expensive, were not very reliable, were constantly in need of maintenance. The first-generation computers examples would include ENIAC, EDVAC, and UNIVAC I.

2nd Generation computers (1956-1963):

The second-generation of computers, or 2nd generation computers, replaced the vacuum tubes of the first-generation computers with transistors. Transistors are smaller, faster, and more reliable than vacuum tubes. The 2nd generation of computers operated using less power than first-generation computers, and utilized a lower amount of energy than that required by first-generation computers to keep the computers cool based on their high-volume circuits. The development of high-level programming languages such as COBOL and FORTRAN took place during this period. Advantages of 2nd generation computers include increase in processing speed, decreased size, decreased cost and an increase in the reliability of 2nd generation computers. The major disadvantage of 2nd generation computers was that the computers still required cooling systems and were relatively expensive. Examples of 2nd generation computers include IBM 1401, IBM 7094.

Third Generation Computers (1964 - 1971):

For the third-generation computers, integrated circuits (ICs) allowed many transistors to be located on one chip. With this advancement, computers were able to process information much more quickly than earlier generations and were also much smaller and less expensive than previous versions of computers. With the development of operating systems and keyboards, computers were also much easier to use than in earlier generations. There were many advantages of third-generation computers such as increased efficiency, decreased maintenance costs, and improved performance over generations one and two. However, the IC technology used to create third-generation computers was very complex and required sophisticated manufacturing techniques. Example of a third-generation computer include IBM Systems/360 and PDP-8.

Fourth Generation Computers (1971 - Present):

Fourth generation computers are Micro-processors, which consist of thousands of integrated circuits soldered onto one chip; thus allowing a computer to operate as an individual computer, instead of being a component in a multi-computer system, as with third generation computers. Micro-processors enabled the development of personal computers, which can be purchased for personal use today. Personal computers are very compact, affordable, and dependable; they also offer fast processing speeds, large storage capacities, and the capability to operate advanced software programs. Although they require routine updates, and can be subject to security breaches through hacking methods, the overall advantages of personal computers are well worth the disadvantages. Example of personal computers are IBM PC, Apple Macintosh and today's models of personal desktops.

Fifth Generation Computing (currently and in the future) :

The fifth generation of computers are focused on Artificial Intelligence (AI) machine, and advanced computing technology. Fifth-generation computer applications will mimic human-like intelligence & have the capability of natural language processing (NLP), robotics & expert systems. Some advantages of fifth-generation computers consist of intelligent decision-making, automated processes, and improved user interactions. Some disadvantages are the high costs of development & concerns for privacy and ethical issues. Some examples of 5th-gen computer applications are AI-based systems like ChatGPT, intelligent robots, and modern-day supercomputers.

Q2.

Storage devices are computer hardware that save data, info, or programs on a computer for use when needed. They save data forever (permanent) as well as temporarily (short-term). They are an essential part of any computer because they help to store an operating system, applications, documents, pictures, movies, and the many files you create.

Storage devices are usually divided into two categories, primary storage and secondary storage. A type of secondary storage device, optical storage devices store data on a disk and use laser devices to read and write data.

Optical Storage Devices:

Optical storage devices are devices that use disks to store data and use laser devices to read the data. They are portable, easy to use and can store a lot of data.

Optical storage devices come in three different types:

1. **Compact Disc (CD)** - Compact Discs were one of the first optical storage devices. They can hold up to 700 MB of data. CDs are used to store music, software, and documents. They are inexpensive and easy to carry. CDs are read using lasers.

The **advantages** of a CD:

- Very portable and durable.
- It is easy to deliver software and media file.

Limitations of a CD:

- Compared to modern storage devices, the capacity of a CD to hold data is limited.

2. **Digital Versatile Disc (DVD)** - DVDs are an enhanced form of CDs with increased potential of storage capability. Storage sizes vary between 4.7 GB (single layer) to 17 GB (dual layer). Primarily used to store audiovisual content and to store computer programs as well as backup to computers.

Advantages:

- Larger storage capacity than CDs
- Higher quality when storing video/audio files

Limitations:

- Less speed than today's flash drives and solid-state drives.

3. **Blu-ray Disc** - Blu-ray Discs are the next generation of optical storage designed for high-definition video and larger file sizes. 25 GB of native storage on a single layer or 50 GB or more for double layer. Higher density of data on each layer: Blue laser versus red laser

Advantages:

- Very high storage capacity
- Ideal for high-definition or 4K video storage

Limitations:

- Greater expense versus CDs and DVDs
- Requires compatible Blu-ray drive.

Q3.

A number system is a way of representing numbers with a defined set of symbols or digits.

Different types of number systems are used by computers to process and store data. The four types of number systems that are most used are binary, decimal, octal, and hexadecimal. These systems all differ based on the value of their base, the symbols used to represent numbers, and how they are used.

1. **The Binary Number System** has a base of 2 with only 2 digits: 0 and 1. Examples of binary numbers are (10), (11), and (100).

The following are the **applications** of the binary number system:

- Inside computers and digital devices.
- In electronic circuits, it represents ON/OFF states.
- It is the basis for all of the processing that computers do.

The **Advantages** of a Binary Number System.

- The binary number system is very simple and well suited for digital equipment.
- It can easily be implemented in electronic circuits.

2. **The Decimal Number System** has a base of 10, using ten digits ranging from 0 to 9. Examples of decimal numbers are (2), (100), and (786).

The following are the **applications** of the decimal number system:

- To perform daily calculations.
- To count and perform mathematical computations.

The **Advantages** of a Decimal Number System

- The decimal number system is easy for people to understand and use.
- This is the most widely used numeric system ever created.

3. **Octal Number System:** The base of the octal number system is eight, and the digits representing numbers in octal run from zero to seven. 17 octal = 15 decimal; 25 octal = 21 decimal; 345 octal = 229 decimal.

The primary **applications** of the octal number system today are:

- To make numbers shorter than are represented in binary form.
- It is commonly used in digital circuitry for computers and other equipment.

Advantages:

- The octal number system is simpler to read because there are fewer digits than the equivalent length of binary.
- Each octal digit will represent three binary digits.

4. **Hexadecimal Number System:** The base of the hexadecimal number system is sixteen, and within the hexadecimal system, 0–9 are used for the numerals zero through nine and letters A–F are used for the following numerals, A = 10, B = 11, C = 12, D = 13, E = 14, and F = 15. Example hexadecimal numbers: 2A, FF, 1B3

The primary **applications** of the hexadecimal number system are:

- some programming languages to represent information within memory.
- used for color representation in web design.

- used by programmers who debug program-related problems.

Advantages:

- In addition to helping carry a large number when represented as a binary number, the hexadecimal number system provides for better comprehension for programmers while coding/decoding the binary representations.

Q4.

The De Morgan Laws represent two important principles of Boolean Algebra that can aid in the simplification of many more complex Boolean Expressions. The laws are generally used throughout Digital Electronics, as they allow us to convert from one logic gate to another, or simplify logical expressions that are complicated.

The two De Morgan Laws are:

The 1st De Morgan Law

The complement of the sum of two variables is equal to the product of the complements.
 $(A+B)' = A'B'$

The 2nd De Morgan Law

The complement of the product of two variables is equal to the sum of the complements.
 $(AB)' = A' + B'$

It is important to note that the De Morgan Laws can also be applied and extended to more than two variables.

Uses of the De Morgans Laws:

- To simplify many more complex Boolean Expressions.
- To design larger Digital Circuits with fewer Logic Gates.
- To convert from AND Gate Style Logic to OR Gate Style Logic (or vice versa).
- To reduce the overall cost and complexity of Digital Circuits.

EXAMPLE:

To get the complement of this Boolean Expression: $(A+B)'$ we can apply the 1st De Morgan Law: $(A+B)' = A'B'$

In other words, we can complement each variable and change our OR Sign to an AND Sign to get the same result as finding the entire Expression's Complement.

Another example of applying the DeMorgan Laws can be seen in looking for the complement

of this Boolean Expression: $(AB)'$

The 2nd De Morgan Law states that the complement of $(AB)' = A' + B'$

Q5.

A **K-Map** is a method that simplifies Boolean functions through graphical representation. The use of a K-Map not only reduces the number of variables used in a digital circuit but also reduces the total number of logic gates required in the circuit as well. A K-Map will show certain input combinations that can never occur or have output values that are irrelevant. These conditions are known as **Don't-Care Conditions** and are represented by "X" in a K-Map.

Don't-care conditions give you additional flexibility while simplifying Boolean functions by allowing you to treat those inputs as either 1 or 0, based on which option will produce a simpler solution.

Guidelines for using don't-care conditions in K-Maps.

1. Use "X" to represent Don't-Care Conditions.

"X" is used on a K-Map to represent the Don't-Care Condition. The "X"s indicate that the output does not matter for those combinations.

2. Use "X" only when it will assist in simplification.

The "X" cell(s) should only be used to create larger groups that will assist with simplifying the Boolean expression. If the "X" cell(s) does not help simplify the Boolean expression, they can be ignored.

3. Create the largest groups possible.

When grouping together the 1s, be sure to include any adjacent "X" cells to create as large a grouping of 2, 4, 8, or 16 cells as possible. The larger the groups, the simpler the resulting Boolean expression.

4. Don't Group Zeros

When you create groupings of ones to satisfy don't care conditions, you must never use the don't care term to create an expression with the same or higher complexity than the original.

5. Follow the Standard K-Map Rules

Standard K-Map rules still apply:

- Groups can consist of 1, 2, 4, 8, or 16 cells.
- Groups must be formed in rectangular shapes.
- Groups may overlap to help simplify the circuit.
- Groups may wrap around from one side of the K-Map to the other.

Example

For a designed Boolean function that has some "1's" and some "X's" in its cells.

If there are 4 "1's", then we could form one group of those 4 (thus allowing us to have fewer variables in our final expression) but there is also an "X" that can be included in the adjacent cell that would form a second group of 8.

In this instance, we would select the larger grouping (the 8) because it will result in simpler expressions and fewer variables in the final expression than if we simply used an "X".

For this case, the don't care condition is treated as a "1" because of its help in simplifying the expression. If it doesn't add to the size of the group, we can treat it as a "0" and not include it.

The Benefits of Using Don't-Care Conditions

- Results in simpler Boolean expressions.
- Reduces the number of logic gates needed.
- Reduces the cost and complexity of the circuit.
- Increases the efficiency and performance of the circuit.

Q6.

A **shift register** is a type of digital circuit called a sequential circuit that stores and transmits binary numbers. It contains several connected flip-flops, each containing one bit of data and having a clock signal to operate. A shift register will **transfer data from one flip-flop to another** in a specific direction.

The name "shift register" indicates how data is sent from the register, i.e., it can be shifted left or right by "shifting" each time there is a clock pulse (or trigger) applied to the register.

An Example of a Shift Register.

A shift register is a collection of several flip-flops connected to each other so that the final output of one flip-flop is the input of the other flip-flops in the series. Each time a clock pulse is applied to a shift register, the contents of the flip-flops will advance by one position moving towards the next flip-flop in the direction of shifting.

If we take as an example a 4-bit shift register containing 2 bits of data (1011) at the time of the first clock pulse, we will see how the contents will be shifted as per the direction in which it is transferred.

Different Types of Shift Registers

1. Serial-In Serial-Out (SISO)

- 1 bit of data comes in and 1 bit of data goes out.
- Used for simple transferring of data.

2. Serial-In Parallel-Out (SIPO)

- Data enters serially but can be read by all outputs simultaneously.
- Used to convert serial data into parallel data.

3. Parallel-In Serial-Out (PISO)

- Data comes into the circuit at the same time (parallel) and leaves the circuit at 1 output pin (serial).
- Used to convert parallel data into serial data.

4. Parallel-In Parallel-Out (PIPO)

- Data enters and leaves the circuit at the same time (parallel).
- Used for temporary data storage.

The shift registers are digital devices used primarily to convert and temporarily store, move or transfer data between devices.

Therefore, they are utilized extensively throughout digital systems for such **applications** as:

- Transient data storage
- Converting Serial data types to Parallel data types
- Converting Parallel data types to Serial data types
- Data transfer between devices
- Process and communicate data
- Provide a Time-Delay in Circuit Operations

Advantages of Shift Registers

- Easy and dependable data storage
- Easy data transfer between circuits
- Helpful in communications and Microprocessor Applications
- Will greatly reduce the amount of complicated wire runs required.